Identification of an unknown- B

Alyssa Barrett

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Introduction

The purpose of this lab is to identify the molecular formula and structure of an unknown compound by using results for NMR, FTIR, and GCMS testing. These various spectrums show signals given off by different kinds of atoms and structures within a molecule. As will be discussed in this paper, Nuclear Magnetic Resonance spectroscopy, or NMR spectroscopy, is a nondestructive method of determining a compound's structure using a magnetic field. Fourier Transform Infrared spectroscopy is a test in which infrared radiation is passed through the sample and a spectrum is created based on where the radiation is absorbed or transmitted (passed through). The final test used is Gas Chromatography Mass Spectroscopy, or GCMS (from here on known as MS) is a technique that is based on vaporizing the substance, and separating out the components of the gas. The mass spectrometer is then used to determine the molecular weight of the compound, which can be coupled with the results from the previous two tests to identify with a high level of certainty the formula and structure for the unknown compound used in the experiment.

It is certainly important when entering into any lab to understand not only the identity of a substance that one will be working with during the lab, but to be familiar with the chemical and physical characteristics and properties of the substance. This will help throughout the lab to understand why the molecules react and interact in a specific way. For example, a ketone is going to behave differently from an alkene in most circumstances, and oftentimes the behavior of one will be favored in a reaction compared with the behavior of the alternative. Therefore, being aware of that difference in structure and behavior will greatly impact the kinds of reactants that someone would want to use in a lab in order to produce a specific reaction or product. Additionally, certain reactants can be particularly hazardous, so mixing in other reactants haphazardly or heating the reaction without understanding the potential hazards of doing so can create an especially dangerous lab environment. Therefore, it is even more important to know the identities and characteristics of the reactants being used.

NMR Spectroscopy

Modern technology allows researchers to employ a technique known as pulsed Fourier Transform Nuclear Magnetic Resonance spectroscopy (FT NMR). In this technique, all of the H or C nuclei are excited by electromagnetic radiation, and the constant decay back to each atom's unexcited state is recorded on a spectrum called a free-induction decay (FID). Each individual signal makes the FT NMR spectrum slightly more complex.

When it comes to interpreting NMR spectrums that are produced, there are four primarily important aspects to note: number of different kinds of protons, chemical shifts, integration, and spin-spin splitting. The number of groups of signals signifies the number of different types of protons in the molecule. For example, a carbon with a carbonyl group, an ethyl group, and an OEt group will show up as three separate groupings on the NMR spectrum. Chemical shifts refer to where the signal groups are on the spectrum, which represents the strength of the signals produced by different protons. For example, electronegative atoms move the signal further downfield (to the left) and unsaturated groups move the signal to the right or left depending on whether the nucleus is in the plane of the double bond. This can be used to help identify many types of functional groups. An important note here is that there may be a shielding effect from electrons surrounding the nucleus from the magnetic field, thus affecting the signal that is produced. If the electron cloud is decreased by adding a halogen group for example, the nucleus is deshielded and is more susceptible to the magnetic field, so a greater chemical shift will occur. Integration is used for determining how many protons are present in various groups. By determining the height ratio of the various peaks in the spectrum, it can be determined how many protons are present in a specific group. Combining this information with the information provided by the chemical shifts, the types of functional groups on the central carbon atom can be determined. Finally, spin-spin splitting, refers to how many peaks there are in a given signal group on the NMR spectrum, and result from the interaction between that nucleus and the adjacent nuclei. This is helpful in arranging the groups that are known to be present in the

molecule. For example, a group with three peaks is known to be adjacent to other nuclei that have a total of N-1,in this case 2, hydrogens attached.

FTIR Spectroscopy

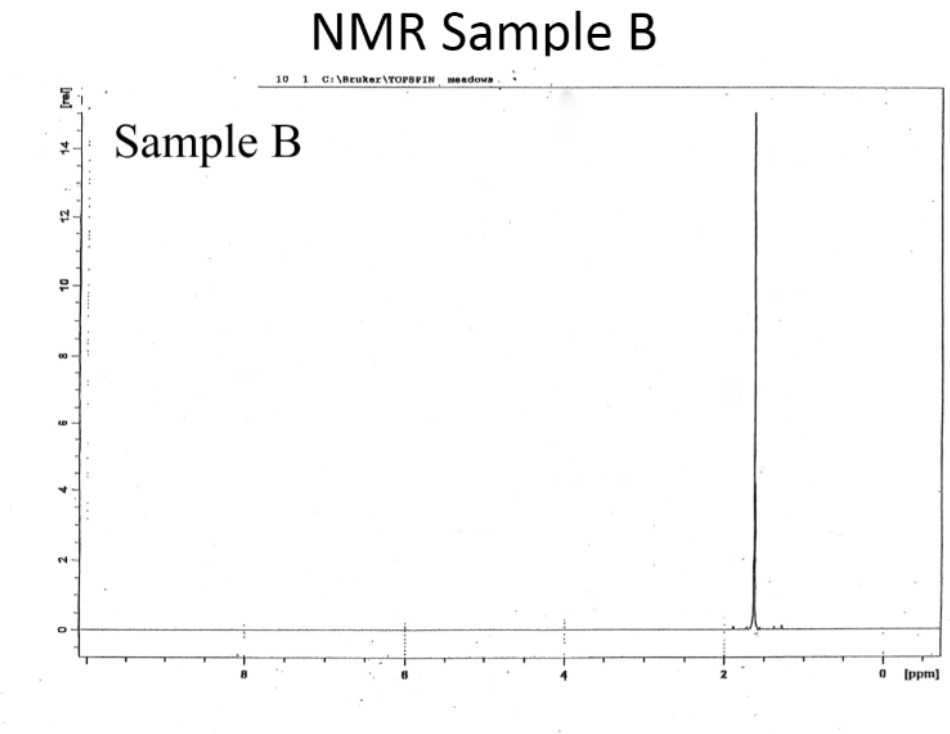
Fourier Transform Infrared Spectrometry (FTIR) is a tool for helping to identify various simple functional groups. In this technique, infrared energy is passed through a sample of the molecule, and bands of absorption are produced. The frequency at which the IR energy is absorbed is directly related to the vibrational energy between atoms in the molecule, in that the IR radiation can only be absorbed when an atom's vibrational energy is at the same frequency as the radiation. Three types of vibration are most important to note for this case, including symmetric stretching vibration, asymmetric stretching vibration, and deformation or bending vibration. FTIR is used by creating two separate spectra and combining them in a way that is useful for interpretation. The first spectrum is a reference spectrum created by passing IR waves through no sample. This helps account for "noise" such as carbon dioxide in the air. The second spectrum is created by passing IR waves through the sample of the molecule being studied. By dividing the latter by the former, an accurate spectrum is produced that is useful for interpretation and comparison. The spectrum produced is often referred to as a chemical fingerprint, meaning that no two molecules have the same spectrum. Using a list of known frequencies and comparing them with the given spectrum, certain functional groups can be determined. In addition to the vibration patterns the strength of the bond connecting two atoms, the masses of the atoms, and the electronegativity of the atoms may all play a part in the spectrum that is produced. By checking for specific strong peak locations, or the lack of a peak in certain locations, the functional groups present in the molecule can be narrowed down. Combining this information with the information provided by the NMR spectroscopy results can almost certainly identify the molecule.

GCMS

Mass spectrometry (MS) is a common technique used in research to determine the molar mass of a compound. In this case, this step is helpful for confirming the results that are produced by the NMR spectrum and the FTIR spectrum. The first step in MS is to ionize the compound, which is generally done by Electron-Impact Mass Spectrometry (EI-MS), which uses electron impact to ionize a sample, and then sorts the ions using a magnetic sector mass analyzer. The ions are sorted by their ration of mass to charge (known as m/z). The detector then analyzes the ions to produce a spectrum of masses via a computer program. GC-MS specifically, called gas chromatograph mass spectrometry combines the described technique of mass spectrometry with a gas chromatography. Often, the spectrum produced will not just show the mass of the entire compound, but will break it down into ions and their relative abundances, creating something of a fingerprint for various compounds.

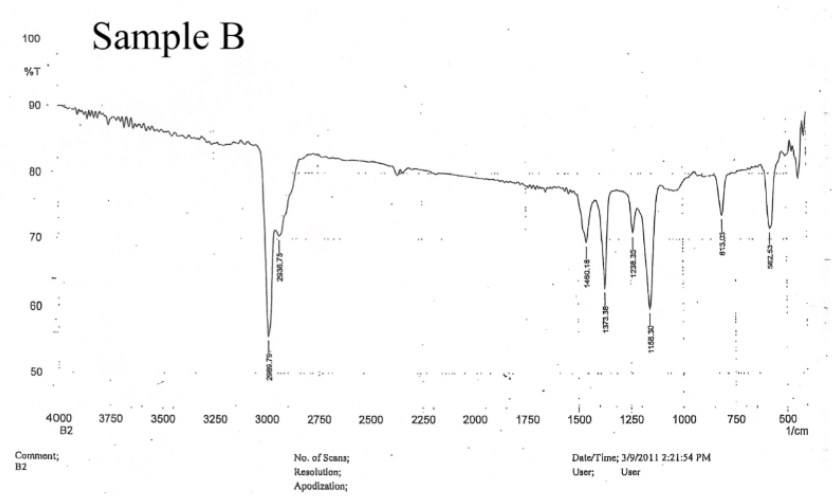
Gas chromatography is used prior to the mass spectrometer and is used to vaporize the compound sample and separate the sample into various components by passing it through a column coated with the stationary phase of that molecule. The column then passes through a heated line and into the mass spectrometer. Working together, these two tools are very useful for idetifying and confirming the molecular makeup of unknown compounds. Much like the two techniques that have already been discussed, namely NMR and FTIR, the spectrum that is produced acts somewhat like a fingerprint, and is often very unique to a specific kind of molecule, maiking it easy to identify what the compound is using this information. Although MS is not quite as precise as the other two, it is still useful in combination with the other two for creating a holistic set of data for the molecule during the identification process. One helpful note is that for hydrocarbons, m/z for the peak can be divided by thirteen to roughly calculate the total number of carbons in the compound. The relationship between peaks of various intensities can be helpful for identifying various groups the the loecule as well, which will be discussed in the analysis portion in more detail.

Sample Data and Analysis



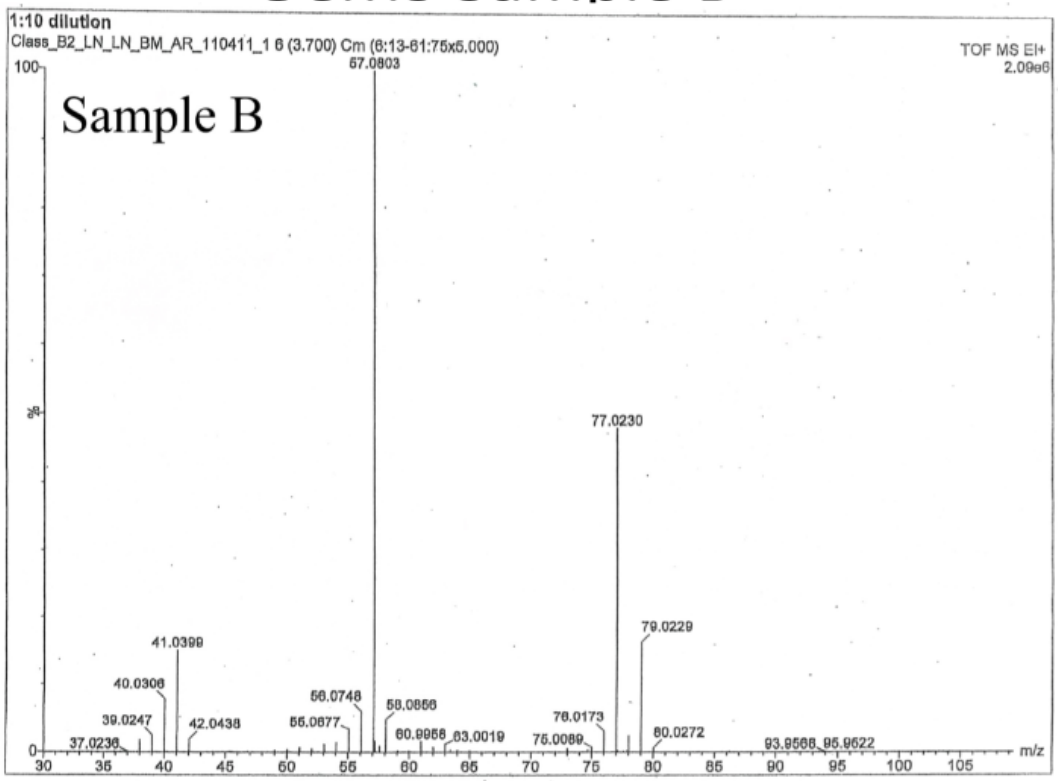
In the course of NMR testing, only one strong signal was produced by the unknown compound. Since the signals primarily tell about the neighbor to neighbor interactions of molecules, this tells us that there is a central carbon that is interacting with its respective neighbors, but it is also the only alpha carbon in the compound. Therefore, if the central carbon is connected to any other carbons, then one, they otherwise only contain hydrogen atoms and two, the central carbon does not contain any hydrogens of its own. It can also be seen that the chemical shift only shifted the peak to roughly 1.5 ppm, which is a relatively weak signal, which often occurs with a bulkier hydrocarbon group such as tert-butyl attached to some electronegative (electron withdrawing group) because the electronegative atom pulls electrons away from the carbon it is attached to, causing the chemical shift, but can only pull at it weakly, as opposed to a scenario in which the electronegative atom is only attached to a methyl, in which case the chemical shift would be greater.

FTIR Sample B



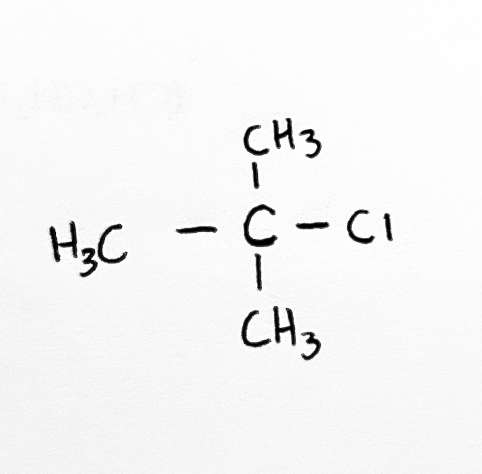
The FTIR spectrum for the sample gives slightly more information about the unknown compound. The first note is the major peak occurring at roughly 2990 cm^-1. This intense peak is a strong indicator of C-H interactions in the compound, specifically stretching the bonds between the carbon and hydrogen, with the carbon being attached to an sp3 hybridized carbon. The 2 less major peaks at around 1153 and 1373, as well as their slightly less neighbors at 1460 and 1238 are strong indicators of the bending vibration of a C-H bond. The final two peaks at 813 and 582 are particularly important because it is relatively infrequent to have a compound produce a strong peak at such a low x position on the spectrum. The lowest of these two is within the prime range for a stretching bond between C and Cl. While it is tempting to use these peaks present in the “aromatic region” of the spectrum to label the unknown as aromatic, it cannot be forgotten that these peaks can also signify unique functional groups such as halogens or other electron withdrawing groups. Aromatic compounds also generally have more presence in that lower space, as opposed to just two peaks as is observed here. Spectra for aromatic compounds also often do not have the strong peak near the 3000 range that is observed in this case. This information, put together, suggests that there are a number of C-H interactions, as well as an sp3 hybridized carbon, and a C-CL bond all present.

GCMS Sample B



The GCMS spectrum is the last piece of the puzzle when it comes to identifying with certainty the provided unknown compound. There are three notable signal groups, occurring respectively around 41, 57, and 77 m/z. Using the conclusions drawn from the previous tests, the MS spectrum can then be observed and compared to what would be expected. The most intense peak present is at 57, with limited presence of isotopes. This could well be expected to represent carbon, since that is supposed at this point to be the most mass-heavy atom in the unknown compound. Dividing 57 by 13, using the “Rule of 13,” the quotient is 4, which checks out for a central carbon atom connected to three other carbons and a halogen as expected. The peak at 77 is another telling, since there is another contribution at 79 that is roughly a third in intensity of the first. This is exactly what would be expected by the presence of chlorine in a compound, so therefore this information confirms what is expected based on the first two spectra.

Conclusion



In conclusion, the identity of the unknown is C(CH3)3Cl, called 2-chloro 2-methylpropane. This is consistent with the NMR spectrum results as the only carbon hydrogen interactions present are the 3 CH3 groups that are attached to a single carbon atom. This explains why there is one intense signal from the NMR reading. As far as the FTIR spectrum, this structure is consistent with the various C-H vibration signals that were picked up, specifically being attached to an sp3 hybridized carbon atom. It also accounts for the strong signal produced by the chlorine atom that is present as the fourth group on the central carbon. Finally, the MS spectrum is consistent, producing the strong signal that represents a monochloride as well as a strong signal from the four carbons present in the molecule.

References

H McNeal. Conversion of Camphene to Camphor. *Chem 1081 Organic Chemistry II Lab Manual.* Macmillan Learning Curriculum Solution: Plymouth, 2023, (p. 85-174).